Agile Process Model: Online Digital Game Marketplace

1. Themes

Theme 1: Marketplace Discovery & Experience

- **Explanation:** Focuses on how users interact with the platform to find, view, and explore content. This encompasses the storefront, search capabilities, and game presentation.
- **Domain Connection:** Directly relates to **Users** browsing **Games** and **Genres**. It ensures the catalog is accessible.

Theme 2: Commerce & Transaction Management

- **Explanation:** Covers the core business logic of exchanging money for digital goods. This includes the shopping cart (if applicable), checkout process, and financial record keeping.
- **Domain Connection:** Manages the lifecycle of **Orders** and **OrderItems**, linking **Users** to the **Games** they purchase.

Theme 3: Ecosystem & Content Supply

- **Explanation:** Focuses on the supply side of the marketplace (Developers) and the management of the entities that populate the store.
- **Domain Connection:** Involves **Developers** creating **Games** and managing **Game** metadata (prices, descriptions).

Theme 4: Identity & Access Control

- **Explanation:** Manages user identities, security, and personal data.
- **Domain Connection:** Handles **User** registration, authentication, and profile data (email, username).

2. Epics

Theme 1: Marketplace Discovery & Experience

- **Epic 1: Storefront & Catalog**
  - **Business Value:** The storefront is the primary entry point. An attractive and functional catalog increases user engagement and conversion rates by showcasing available games effectively.
  - **Related Entities:** Game, Genre.
- **Epic 2: Search & Discovery**
  - **Business Value:** As the library grows, users need efficient ways to find specific content. Good search/filter tools reduce friction in the buying process.
  - **Related Entities:** Game, Genre.

Theme 2: Commerce & Transaction Management

- **Epic 3: Purchasing System**
  - **Business Value:** This is the revenue generator. A smooth, error-free purchasing flow is critical for business viability.
  - **Related Entities:** User, Order, OrderItem, Game.
- **Epic 4: Order History & Library**
  - **Business Value:** Users need to trust that their purchases are recorded. Accessing past orders builds trust and allows users to track their spending/library.
  - **Related Entities:** User, Order, OrderItem.

Theme 3: Ecosystem & Content Supply

- **Epic 5: Developer Game Management**
  - **Business Value:** Without games, there is no marketplace. Empowering developers to easily list and manage their games ensures a steady stream of content.
  - **Related Entities:** Developer, Game, Genre.

Theme 4: Identity & Access Control

- **Epic 6: User Account Management**

    - **Business Value:** Essential for identifying customers, securing transactions, and maintaining long-term user relationships.
    - **Related Entities:** User.

3. User Stories

US-01: User Registration

- **As a** new visitor, **I want** to register an account with my email and username, **so that** I can purchase games and save my history.
- **Acceptance Criteria:**
    - Given a visitor is on the registration page.
    - When they enter a unique email, unique username, and valid password.
    - Then a new User record is created with the current date as registrationDate.
    - And they are logged in automatically.

US-02: User Login

- **As a** registered user, **I want** to log in, **so that** I can access my account.
- **Acceptance Criteria:**
    - Given a user is on the login page.
    - When they enter valid credentials.
    - Then they are authenticated and redirected to the homepage.

US-03: Browse All Games

- **As a** user, **I want** to see a list of all available games, **so that** I can explore the catalog.
- **Acceptance Criteria:**
    - Given the user is on the homepage.
    - When the page loads.
    - Then a list of Games is displayed showing Title and Price.

US-04: View Game Details

- **As a** user, **I want** to click on a game to see its full details, **so that** I can decide if I want to buy it.
- **Acceptance Criteria:**
    - Given a user sees a game in a list.

    - When they click the game title.
    - Then they are taken to a detail page showing Title, Description, Price, Release Date, Developer Name, and Genres.

US-05: Filter Games by Genre

- **As a** user, **I want** to filter the game list by genre, **so that** I can find specific types of games I like.
- **Acceptance Criteria:**
    - Given the user is viewing the game catalog.
    - When they select "RPG" from a genre dropdown.
    - Then only Games associated with the "RPG" Genre are displayed.

US-06: Purchase a Single Game

- **As a** user, **I want** to buy a specific game immediately, **so that** I can own it.
- **Acceptance Criteria:**
    - Given a user is on a Game Detail page.
    - When they click "Buy Now" and confirm payment.
    - Then an Order is created with one OrderItem for that game.
    - And the Order totalAmount equals the Game price.

US-07: View Order History

- **As a** user, **I want** to see a list of my past purchases, **so that** I can verify what I have bought.
- **Acceptance Criteria:**
    - Given a logged-in user.
    - When they navigate to "My Orders".
    - Then they see a list of all Orders linked to their userId, showing Order Date and Total Amount.

US-08: View Order Details

- **As a** user, **I want** to click an order to see what games were in it, **so that** I can recall specific purchase details.
- **Acceptance Criteria:**
    - Given a user is viewing their Order History.

    - When they click on an Order ID.
    - Then they see the OrderItems (Game Title, Purchase Price) for that order.

US-09: Developer Registration

- **As a** game creator, **I want** to register as a Developer, **so that** I can publish my games.
- **Acceptance Criteria:**
    - Given an admin or registration interface.
    - When a new Developer name is entered.
    - Then a Developer entity is created in the system.

US-10: Publish New Game

- **As a** Developer, **I want** to list a new game, **so that** users can buy it.
- **Acceptance Criteria:**
    - Given a logged-in Developer.
    - When they submit a form with Title, Description, Price, and Genres.
    - Then a new Game entity is created linked to this Developer.

US-11: Update Game Price

- **As a** Developer, **I want** to update the price of my game, **so that** I can run sales or adjust for the market.
- **Acceptance Criteria:**
    - Given a Developer owns a specific game.
    - When they change the price field and save.
    - Then the Game's price is updated in the database.

US-12: Search Games by Name

- **As a** user, **I want** to search for a game by its title, **so that** I can find a specific game quickly.
- **Acceptance Criteria:**
    - Given the search bar.
    - When the user types "Cyber".
    - Then games with "Cyber" in the title are displayed.
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US-13: Multi-Game Checkout (Cart)

- **As a** user, **I want** to add multiple games to a cart and checkout once, **so that** I save time.
- **Acceptance Criteria:**
    - Given a user has added 2 games to their cart.
    - When they complete checkout.
    - Then one Order is created with two OrderItems.
    - And the totalAmount is the sum of both game prices.

US-14: View Games by Developer

- **As a** user, **I want** to click a developer's name, **so that** I can see all games they have made.
- **Acceptance Criteria:**
    - Given a user is on a Game Detail page.
    - When they click the Developer's name.
    - Then a list of all Games linked to that DeveloperId is shown.

US-15: Admin Dashboard (System Stats)

- **As an** admin, **I want** to see total sales, **so that** I can monitor platform performance.
- **Acceptance Criteria:**
    - Given an admin user.
    - When they view the dashboard.
    - Then they see the sum of totalAmount from all Orders.
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4. Tasks (Breakdown for Major Stories)

Breakdown for US-04: View Game Details

1. **Task:** Create Game and Developer database tables/entities with foreign keys.
2. **Task:** Implement Backend API GET /api/games/{id} to fetch game details + developer name.
3. **Task:** Create Frontend "Game Detail" component layout.
4. **Task:** Implement logic to fetch data from API on component mount.

5. **Task:** Style the display of Price, Description, and Genres.
6. **Task:** Write Unit Test for GameService.getGameById.

Breakdown for US-06: Purchase a Single Game

1. **Task:** Create Order and OrderItem database tables.
2. **Task:** Implement Backend API POST /api/orders that accepts userId and gameId.
3. **Task:** Implement transaction logic: Calculate total, save Order, save OrderItem.
4. **Task:** Create "Buy Now" button component on Game Detail page.
5. **Task:** Implement success/failure notification UI after purchase attempt.
6. **Task:** Validate that a user cannot buy a game they already own (optional business rule logic).

Breakdown for US-01: User Registration

1. **Task:** Create User table with constraints (unique email/username).
2. **Task:** Implement Password Hashing utility (security).
3. **Task:** Create Backend API POST /api/register.
4. **Task:** Design Registration Form UI (inputs for email, username, password).
5. **Task:** Implement client-side validation (email format, empty fields).
6. **Task:** Integrate form submission with backend API.

5. Product Backlog

ID	User Story Title	Epic	Priority	Story Points
US-01	User Registration	Epic 6: User Account Management	High	5
US-02	User Login	Epic 6: User Account Management	High	3
US-09	Developer Registration	Epic 5: Developer Game Management	High	3
US-10	Publish New Game	Epic 5: Developer Game Management	High	8
US-03	Browse All Games	Epic 1: Storefront & Catalog	High	5
US-04	View Game Details	Epic 1: Storefront & Catalog	High	3
US-06	Purchase a Single Game	Epic 3: Purchasing System	High	8
US-07	View Order History	Epic 4: Order History & Library	Medium	5
US-08	View Order Details	Epic 4: Order History & Library	Medium	3
US-12	Search Games by Name	Epic 2: Search & Discovery	Medium	5
US-05	Filter Games by Genre	Epic 2: Search & Discovery	Medium	5
US-14	View Games by Developer	Epic 2: Search & Discovery	Low	2
US-11	Update Game Price	Epic 5: Developer Game Management	Low	3
US-13	Multi-Game Checkout (Cart)	Epic 3: Purchasing System	Low	13
US-15	Admin Dashboard	Epic 5: Developer Game Management	Low	8

6. Sprint Planning

Sprint 1: Foundation & Catalog

- **Sprint Goal:** Establish the user database and allow users to view the available games catalog.
- **Included Stories:**
  - US-01: User Registration (5 pts)
  - US-02: User Login (3 pts)
  - US-09: Developer Registration (3 pts)
  - US-10: Publish New Game (8 pts)
  - US-03: Browse All Games (5 pts)
- **Total Points:** 24
- **Deliverables:** A working system where users can sign up/in, developers can add games, and the homepage lists these games.

Sprint 2: Commerce & Transactions

- **Sprint Goal:** Enable the core business function: users purchasing games.
- **Included Stories:**
  - US-04: View Game Details (3 pts)
  - US-06: Purchase a Single Game (8 pts)
  - US-07: View Order History (5 pts)
  - US-08: View Order Details (3 pts)
- **Total Points:** 19
- **Deliverables:** Users can click a game, buy it, and see the record of that purchase in their history.

Sprint 3: Discovery & Experience Enhancements

- **Sprint Goal:** Improve usability with search, filtering, and better navigation.
- **Included Stories:**
  - US-12: Search Games by Name (5 pts)
  - US-05: Filter Games by Genre (5 pts)
  - US-11: Update Game Price (3 pts)

    - US-14: View Games by Developer (2 pts)
- **Total Points:** 15
- **Deliverables:** A polished storefront where users can find specific content easily and developers can manage pricing.
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7. PSI (Potentially Shippable Increment)

- **PSI 1 (End of Sprint 1):**
    - A deployed web application where a user can register and log in.
    - A "Developer" mode allows adding games to the database.
    - The Home Page displays a raw list of all added games (Title/Price).
    - *Limitation:* No buying capability yet.
- **PSI 2 (End of Sprint 2):**
    - Full "Buy Now" flow is functional. Money is not actually processed (mocked), but Orders are correctly saved to the database.
    - Users have a "My Profile" area listing their purchased games.
    - Game Detail pages exist.
    - *Value:* The system is now a functional MVP for commerce.
- **PSI 3 (End of Sprint 3):**
    - Search bar works.
    - Genre dropdown filters the list.
    - Developers can edit prices.
    - *Value:* The system is user-friendly and ready for a wider beta release.

8. MVP Definition (Minimum Viable Product)

The MVP for the Online Digital Game Marketplace is defined as the output of **Sprint 1 + Sprint 2**.

- **Included Themes:** Marketplace Discovery, Commerce, Identity, Content Supply.
- **Included Epics:** Storefront, Purchasing, User Accounts, Developer Management.
- **Critical Features:**
    1. User Registration/Login.
    2. Developer ability to add Games.

    3. Public Catalog (List & Details).
    4. Buying a game (Order creation).
    5. Order History.
- **Exclusions (Post-MVP):**
    - Shopping Cart (Multi-game checkout).
    - Advanced Search/Filtering (Basic list is enough for MVP).
    - Admin Dashboard.

9. Trello Board Structure

Board Name: Digital Game Marketplace Development

Lists (Columns):

1. **Product Backlog** (All future stories)
2. **Sprint Backlog** (Selected for current Sprint)
3. **In Progress** (Currently being worked on)
4. **Code Review** (PR submitted, waiting for peer review)
5. **Testing / QA** (Deployed to staging, verifying acceptance criteria)
6. **Done** (Passed QA, ready for release)

Card Structure:

- **Title:** [ID] Title (e.g., [US-06] Purchase a Single Game)
- **Description:**
    - **User Story:** "As a user..."
    - **Acceptance Criteria:** Bulleted list of Given/When/Then.
    - **Technical Notes:** API endpoints, DB tables involved.
- **Labels:**
    - Frontend (Blue)
    - Backend (Green)
    - Database (Yellow)
    - High Priority (Red)
    - Bug (Orange)

Sample Cards:

1. **Card Title:** [US-01] User Registration
    - **List:** Done
    - **Labels:** Backend, Frontend, High Priority
    - **Content:** Implement registration form and API. Ensure email uniqueness.
2. **Card Title:** [US-06] Purchase a Single Game
    - **List:** In Progress
    - **Labels:** Backend, Database, High Priority
    - **Content:** Create Order entity. Logic: Input(userId, gameId) -> Output(Order).
3. **Card Title:** [US-10] Publish New Game
    - **List:** Sprint Backlog
    - **Labels:** Backend, Database
    - **Content:** Form for developers to add Title, Price, Desc. Insert into Games table.
4. **Card Title:** [US-03] Browse All Games
    - **List:** Code Review
    - **Labels:** Frontend
    - **Content:** Fetch list from GET /api/games. Render as grid cards.
5. **Card Title:** [US-12] Search Games by Name
    - **List:** Product Backlog
    - **Labels:** Frontend, Backend
    - **Content:** Add search bar to header. Filter query param in API.